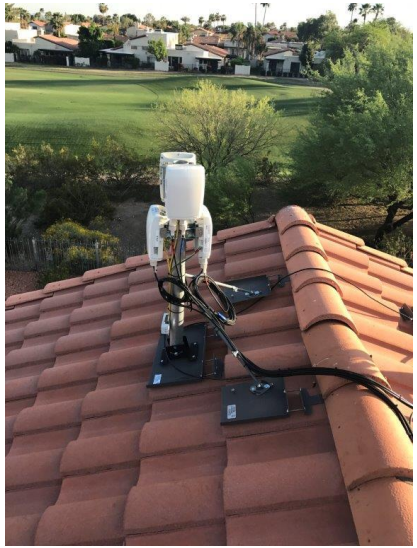
Gigabit to the Home

5G Fixed Wireless Access in mmWave

NZNOG 2019

- Introduction to mmWave
- Gigabit to the home

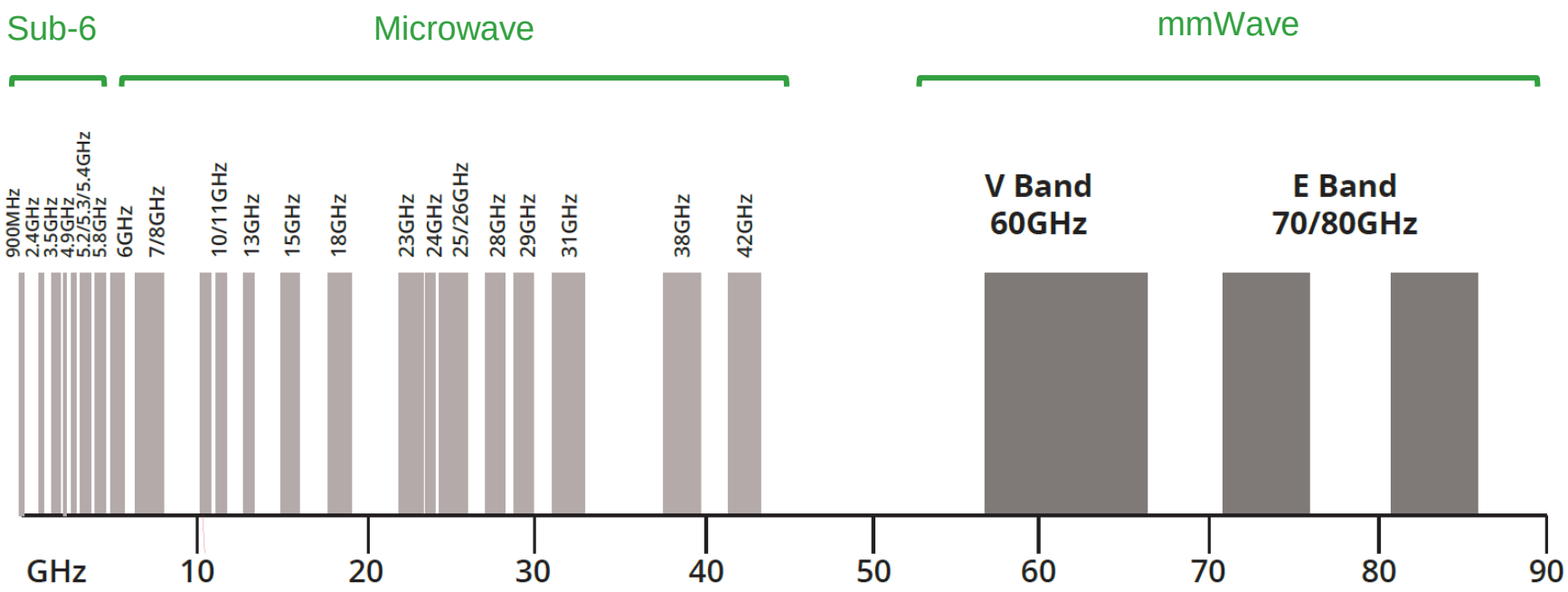


Daniel Ephraty
Director of Sales Engineering
Siklu Communications

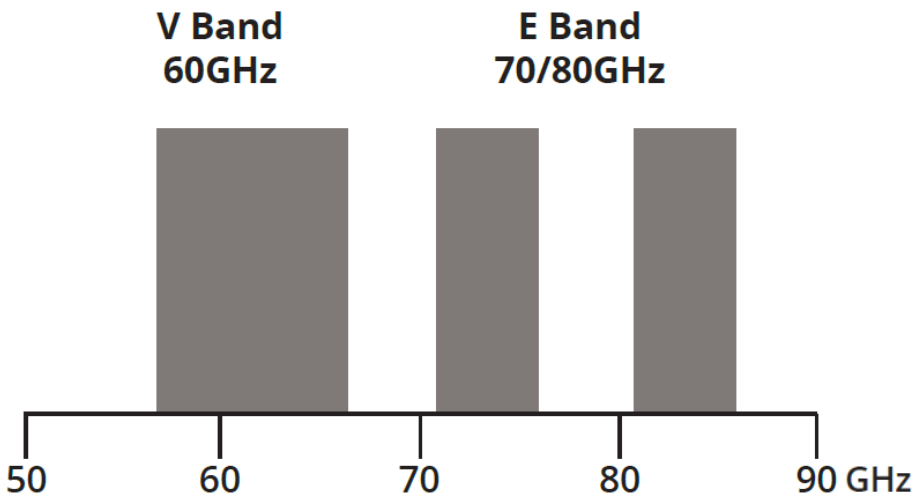
Fixed Wireless Access

- Earlier attempts at FWA
  - Naughties (LMDS,…)
  - WiMAX
- Challenges
  - Performance as good as wireline
  - Comparable or better cost
  - Plenty of spectrum
- mmWave (“5G”) – delivers on these challenges

The Fixed Wireless Spectrum



mmWave in NZ



V-band

- Up to 7GHz: 57-64GHz
- License exempt

E-band

- 10GHz: 71-76GHz, 81-86GHz
- Licensed ($150)

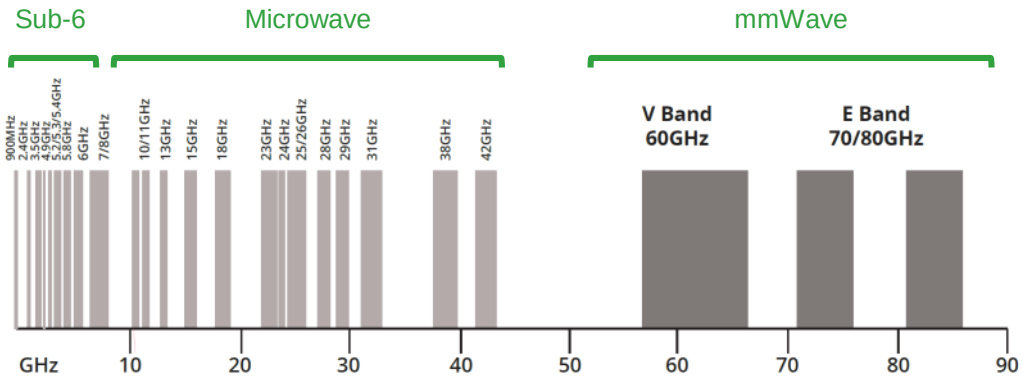
Characteristics of mmWaves

- **Abundant spectrum**
- Narrow beam-widths
- Line of sight
- Link distance

Abundant spectrum

- Uncongested
- Wide channels available (2GHz)

  $\Rightarrow$ Gigabit throughputs (10Gbps)



Characteristics of mmWaves

- Abundant spectrum
- **Narrow beam-widths**
- Line of sight
- Link distances

Narrow beam-widths

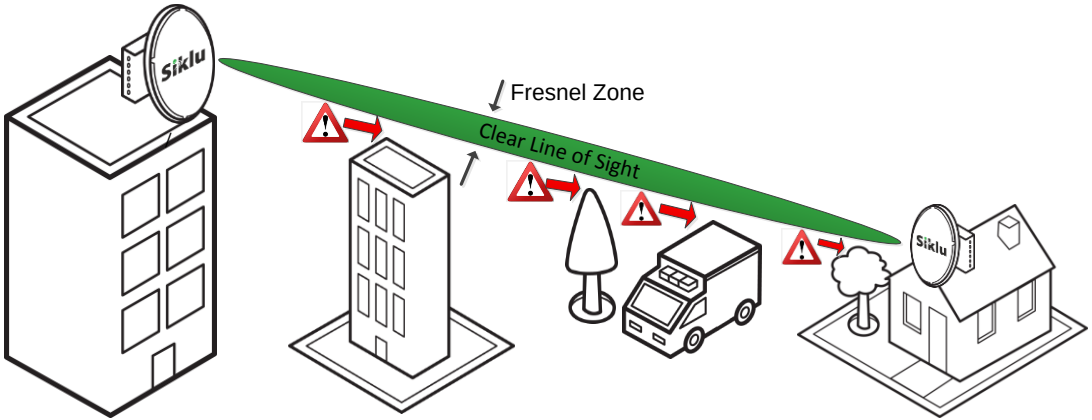
- “Pencil beams” (0.5° to 2.5°)
- No interference
- High frequency reuse
- Secure transmission
- Stability essential when <1°

Characteristics of mmWaves

- Abundant spectrum
- Narrow beam-widths
- **Line of sight**
- Link distances

Line of sight

- Clear line of sight is essential
- Negligible Fresnel zone



Characteristics of mmWaves

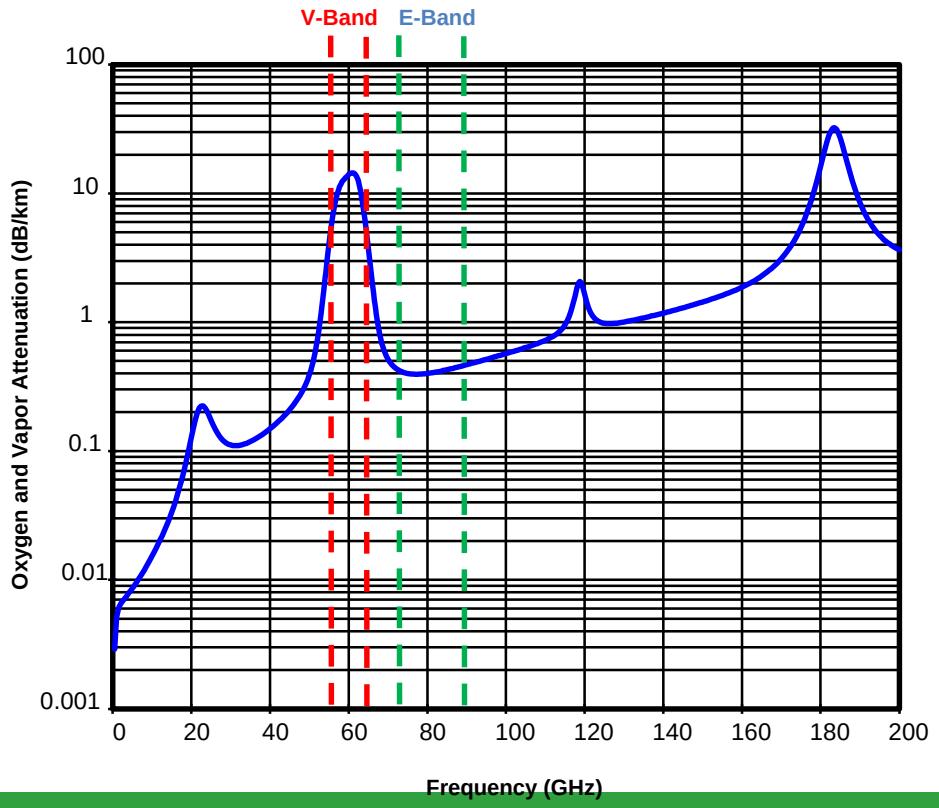
- Abundant spectrum
- Narrow beam-widths
- Line of sight
- **Link distances**

Link Distance

- V-band additionally attenuated by the oxygen in the air
- V-band and E-band attenuated by rain
- Effective distances:
  - V-band: up to ~1Km
  - E-band:
    - up to ~3Km for 10Gbps
    - Longer for lower air capacities
    - Longer with overbuild ~10Km

Atmospheric Absorption

- Oxygen attenuation at 71-76/81-86GHz
  - ~ 0.5dB/Km
  - Negligible effect
- Oxygen attenuation at 57-64GHz
  - Up to 15dB/Km
  - Limited distance, suitable for dense urban areas



Link Design in Millimeter Waves

Factors that count

- Attenuation due to rain
  - Excellent long-term statistics available, e.g., **ERA40**: ECMWF's database based on 45 years of rain data (1957-2002)
- Oxygen absorption (V-band only)

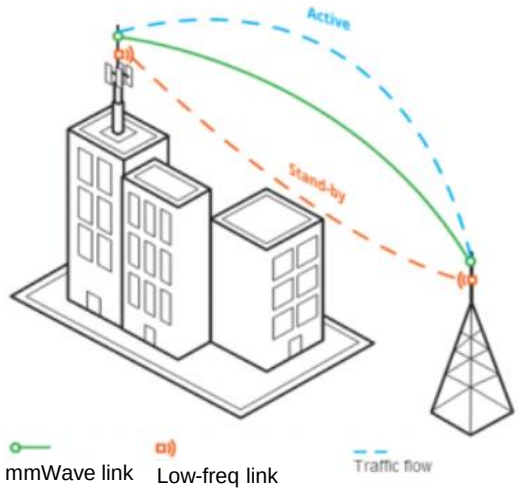
Factors that don't count

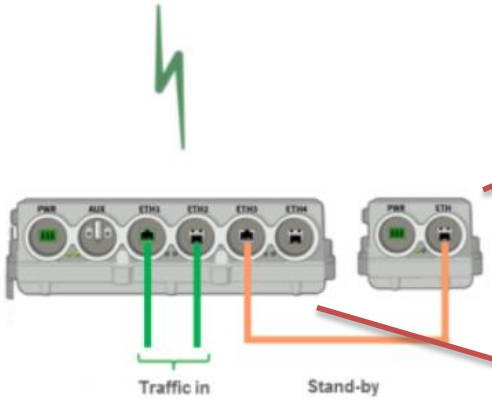
- Interference
- DFS
- Multipath due to surface reflection
  - Attenuation (flat fading)
  - Distortion (selective fading)
  - Depolarization
- Fading due to beam spreading
- Diffraction due to obstructions

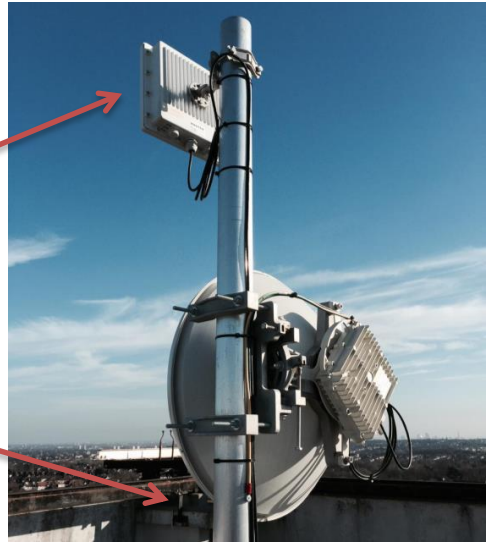
Easy to predict

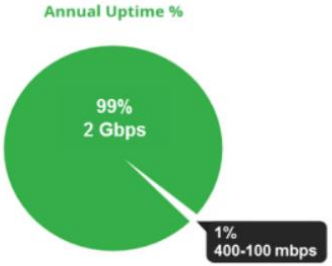
Difficult to predict

Overbuild

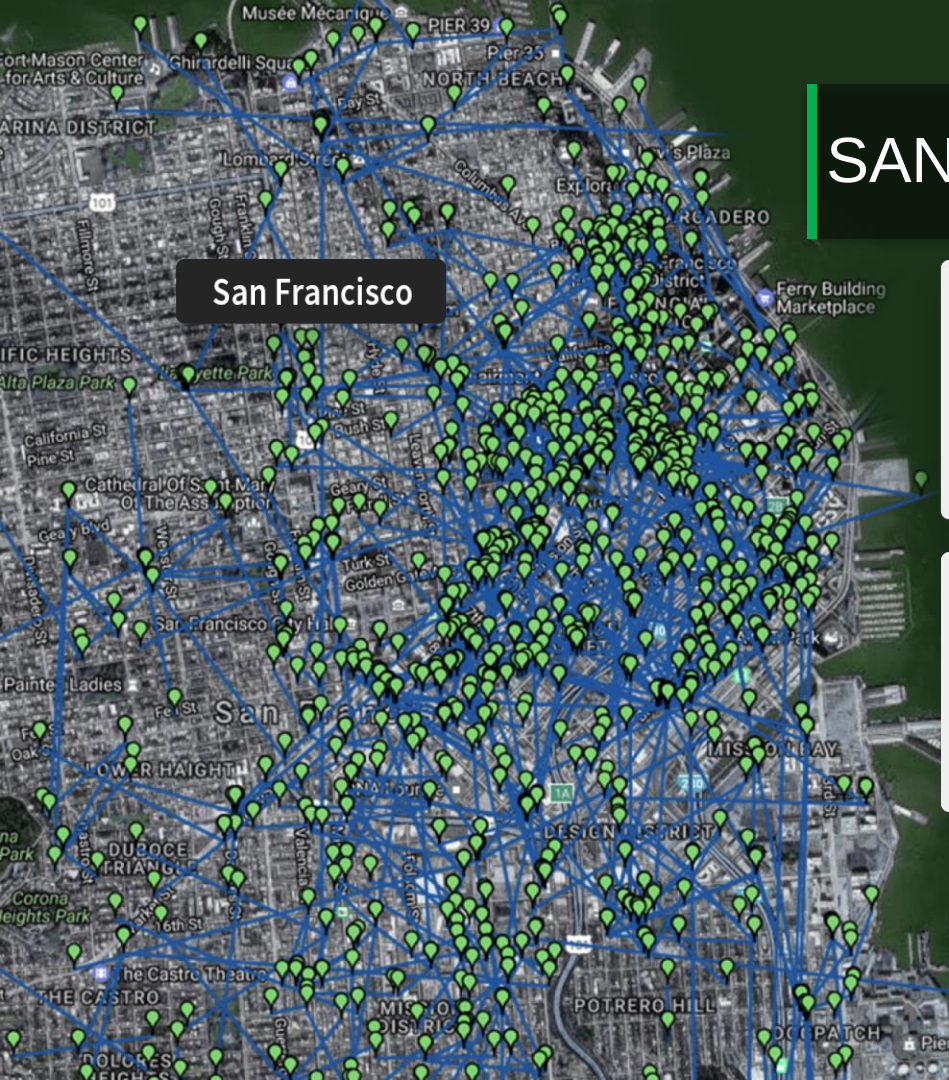

Active
Stand-by
mmWave link
Low-freq link
Traffic flow


Traffic in
Stand-by




Annual Uptime %
99%
2 Gbps
1%
400-100 mbps

SAN FRANCISCO DEPLOYMENTS



Video Surveillance

Public Safety

WiFi Backhaul

Education

Transportation

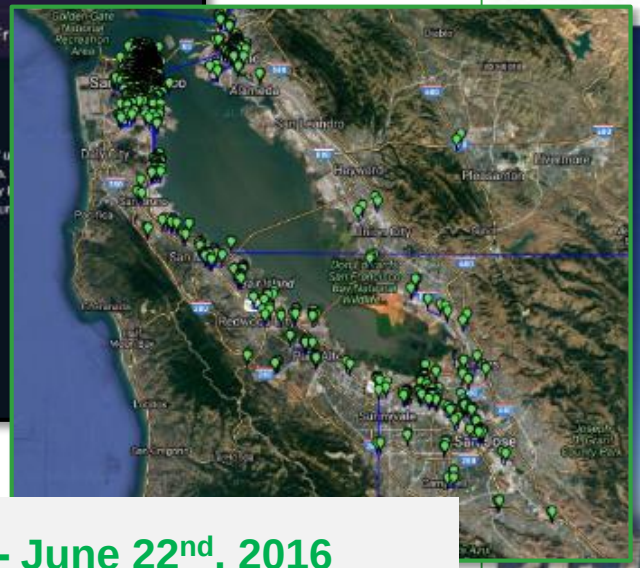
Health Care

Fixed Wireless Broadband

Case in Point: Webpass

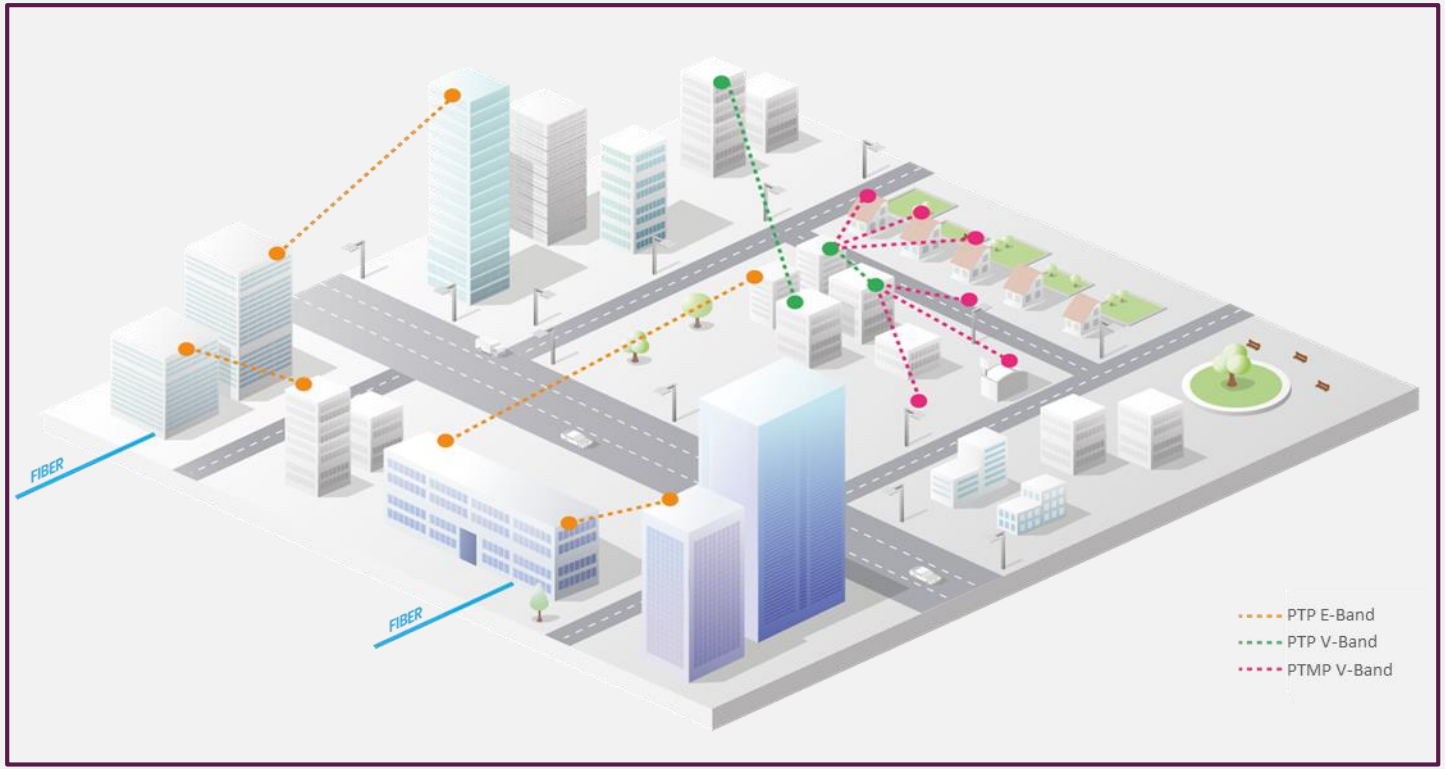
Marking a Milestone: 1000 Customers with 1000 Mbps

Posted February 12 2016 by Webpass



Acquired by Google Fiber – June 22nd, 2016

Wireless ‘Fibre’

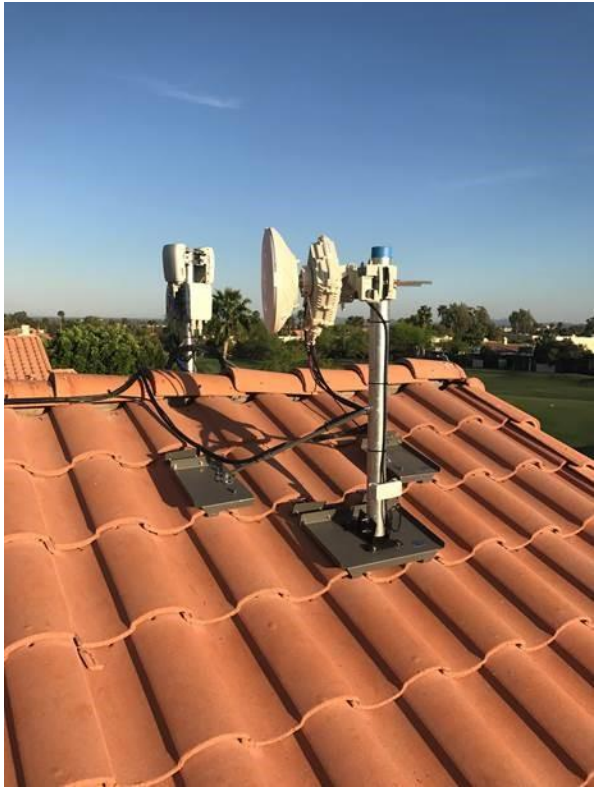


Gigabit to the home

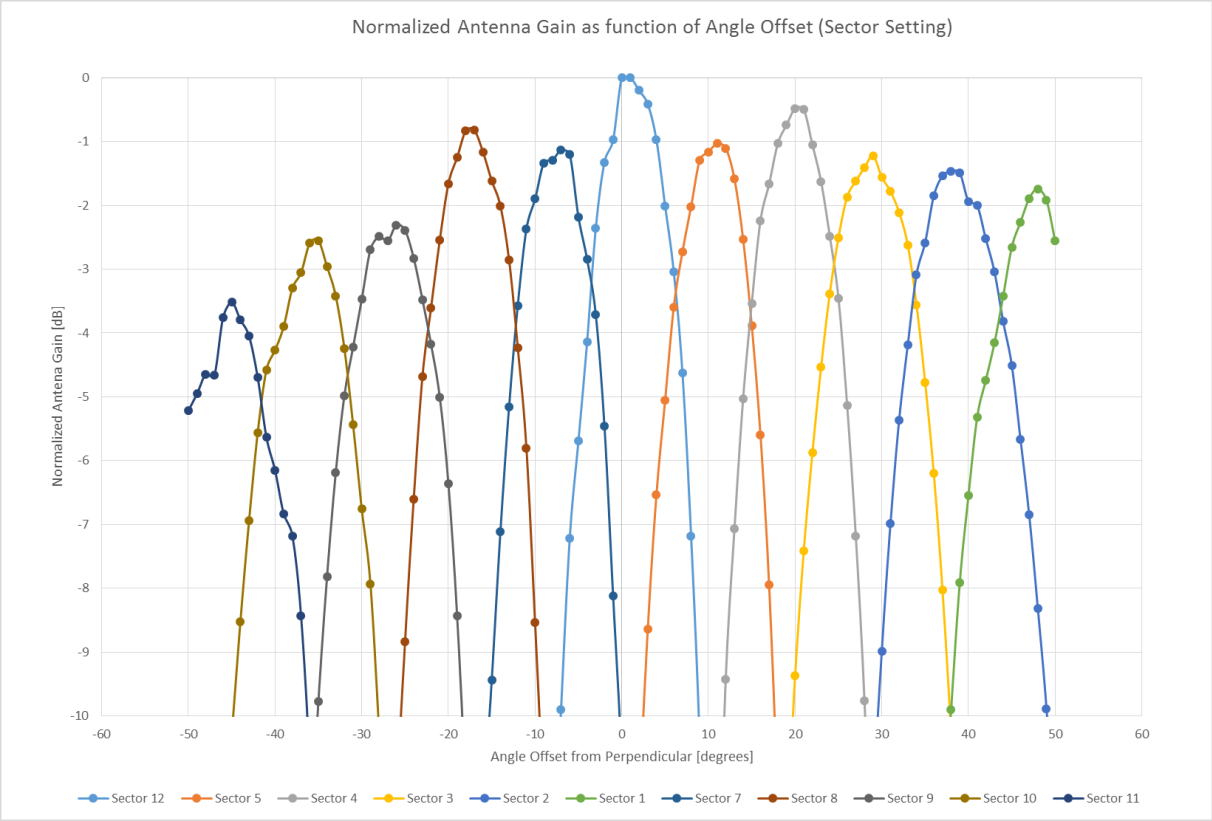
Fibre extension: E-band

Access V-band

- Inexpensive chipset
  - 802.11ad (Wigig)
  - 802.11ay (proposed enhancement)
  - Terragraph
- Small footprint
- Beamforming, simplifies installation – sector coverage
- Topology
  - Ptp with redundancy (rings)
  - Ptmp with redundancy (multiple APs)
  - Ptmp without redundancy

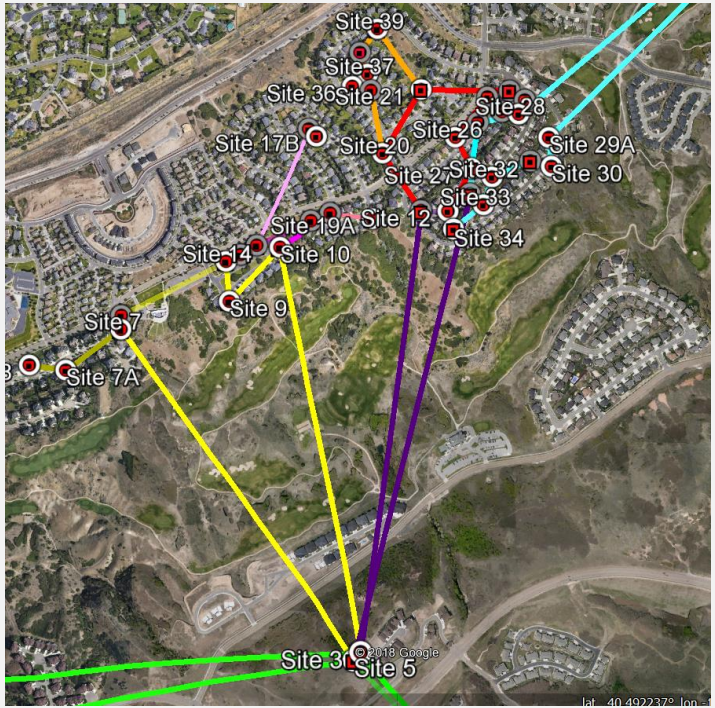


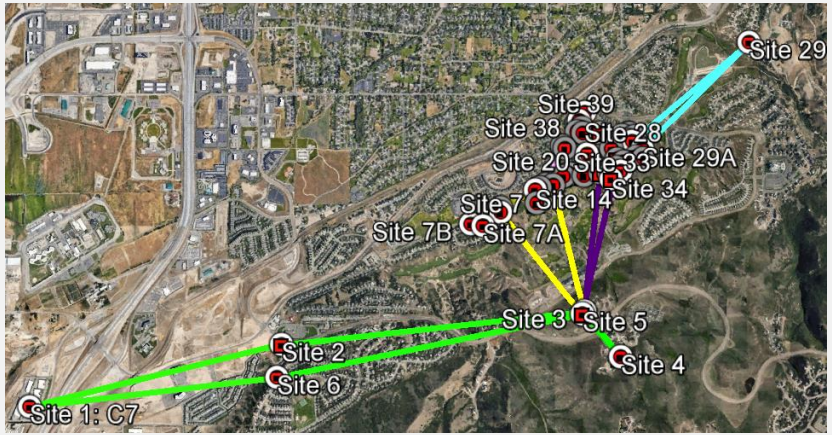
Sector coverage with narrow beams



Design example: Utah, USA

ptp 60GHz with ring protection





Ethernet Ring Protection (ERP), switching time <50ms

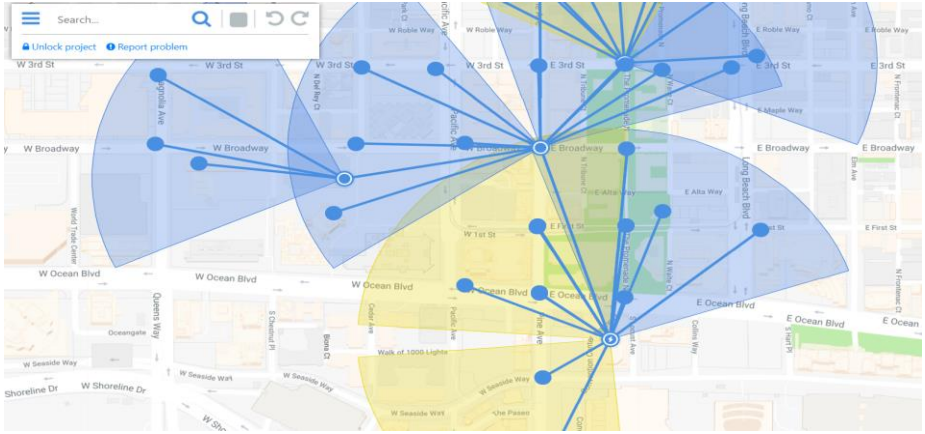
Design example: Georgia, USA

ptmp 60GHz

359/359 homes connected (100%)
16x poles (average 22 homes/pole)
All on fiber route
53x APs sectors (90^0) + 359 terminals (average 7 homes/BU, 3.3 APs/Pole)

Automated planning

- FWA network planning presents challenges
  - ‘Business Logic’
  - Asset location
  - Topology preference
  - Line of sight calculation
  - Link availability calculation
  - Configuration



Q&A

THANK YOU

Siklu